📓 Modul Minggu 5 – Laravel: Migration & Model

🎯 Tujuan Pembelajaran

1. Mahasiswa memahami struktur MVC pada Laravel.
2. Mahasiswa mampu membuat **migration, model, dan seeder**.
3. Mahasiswa dapat menjalankan migrasi database MySQL di Laravel.

📖 Materi Teori

1. Struktur Laravel (MVC)

Laravel menggunakan **arsitektur MVC (Model–View–Controller)**:

- **Model** → representasi data dari tabel di database (misalnya `Book.php`, `Member.php`).
- **View** → tampilan (Blade Template atau frontend Vue.js, nanti dipisahkan).
- **Controller** → penghubung logika bisnis antara model dan view.

Struktur folder utama:

```
app/
   Models/      # Model database
   Http/
       Controllers/
database/
   migrations/   # File migration
   seeders/      # File seeder
```

2. Migration

- **Migration** adalah blueprint (cetak biru) untuk membuat tabel database dengan perintah Laravel.
- Dijalankan dengan:
- `php artisan make:migration create_books_table`
- `php artisan migrate`

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3. Seeder

- **Seeder** digunakan untuk mengisi tabel dengan **dummy data** awal.
- Dijalankan dengan:
- `php artisan db:seed`
-

4. Model

- **Model** merepresentasikan tabel di database.
- Contoh: `Book.php` untuk tabel `books`.
- Dibuat dengan:
- `php artisan make:model Book`
-

🛠 Langkah Praktik

1. Konfigurasi Database di `.env`

Edit file `.env`:

```
DB_CONNECTION=mysql

DB_HOST=127.0.0.1

DB_PORT=3306

DB_DATABASE=digital_library

DB_USERNAME=root

DB_PASSWORD=
```

2. Buat Migration

Buat migration untuk tabel: anggota, buku, author, penerbit, peminjaman.

Contoh:

```
php artisan make:migration create_members_table

php artisan make:migration create_books_table

php artisan make:migration create_authors_table

php artisan make:migration create_publishers_table
```

```
php artisan make:migration create_loans_table
```

3. Isi File Migration

Contoh `create_members_table.php`:

```
public function up()
{
    Schema::create('members', function (Blueprint $table) {
        $table->id();
        $table->string('name');
        $table->string('email')->unique();
        $table->string('phone');
        $table->timestamps();
    });
}
```

Contoh `create_books_table.php`:

```
public function up()
{
    Schema::create('books', function (Blueprint $table) {
        $table->id();
        $table->string('title');
        $table->foreignId('author_id')->constrained()->onDelete('cascade');
$table->foreignId('publisher_id')->constrained()->onDelete('cascade');
        $table->timestamps();
    });
}
```

Contoh `create_loans_table.php`:

```
public function up()
{
    Schema::create('loans', function (Blueprint $table) {
        $table->id();
        $table->foreignId('member_id')->constrained()->onDelete('cascade');
        $table->foreignId('book_id')->constrained()->onDelete('cascade');
        $table->date('loan_date');
        $table->date('return_date')->nullable();
        $table->timestamps();
    });
}
```

4. Jalankan Migration

```
php artisan migrate
```

5. Buat Model

```
php artisan make:model Member
php artisan make:model Book
php artisan make:model Author
php artisan make:model Publisher
php artisan make:model Loan
```

6. Buat Seeder

```
php artisan make:seeder MemberSeeder
php artisan make:seeder BookSeeder
```

Contoh `MemberSeeder.php`:

```
public function run()

{

    DB::table('members')->insert([

        ['name' => 'Ali', 'email' => 'ali@mail.com', 'phone' => '081234567'],

        ['name' => 'Budi', 'email' => 'budi@mail.com', 'phone' =>
'081987654'],

    ]);

}
```

Jalankan:

```
php artisan db:seed --class=MemberSeeder
```

📌 Tugas / Milestone

1. Buat migration untuk semua tabel (`members`, `books`, `authors`, `publishers`, `loans`).
2. Buat minimal 1 seeder untuk mengisi dummy data.
3. Pastikan **migration & seeder berjalan lancar** di MySQL.
4. Push kode ke repository GitHub.

file jadi siap copy-paste untuk migration & seeder Laravel sesuai tabel:

🗁 Struktur project:

```
database/
   ├── migrations/
   │      ├── 2025_xx_xx_create_members_table.php
   │      ├── 2025_xx_xx_create_authors_table.php
   │      ├── 2025_xx_xx_create_publishers_table.php
   │      ├── 2025_xx_xx_create_books_table.php
   │      └── 2025_xx_xx_create_loans_table.php
   └── seeders/
          ├── MemberSeeder.php
          ├── AuthorSeeder.php
          ├── PublisherSeeder.php
          ├── BookSeeder.php
          └── DatabaseSeeder.php
```

◈ Migration Files

1. `create_members_table.php`

```
<?php


use Illuminate\Database\Migrations\Migration;

use Illuminate\Database\Schema\Blueprint;

use Illuminate\Support\Facades\Schema;


return new class extends Migration {

    public function up(): void {

        Schema::create('members', function (Blueprint $table) {
```

```
            $table->id();
            $table->string('name');
            $table->string('email')->unique();
            $table->string('phone');
            $table->timestamps();
        });
    }

    public function down(): void {
        Schema::dropIfExists('members');
    }
};
```

2. `create_authors_table.php`

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration {
    public function up(): void {
        Schema::create('authors', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->timestamps();
        });
    }
```

```
    public function down(): void {
        Schema::dropIfExists('authors');
    }
};
```

3. create_publishers_table.php

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration {
    public function up(): void {
        Schema::create('publishers', function (Blueprint $table) {
            $table->id();
            $table->string('name');
            $table->timestamps();
        });
    }

    public function down(): void {
        Schema::dropIfExists('publishers');
    }
};
```

4. create_books_table.php

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration {
    public function up(): void {
        Schema::create('books', function (Blueprint $table) {
            $table->id();
            $table->string('title');
$table->foreignId('author_id')->constrained()->onDelete('cascade');
$table->foreignId('publisher_id')->constrained()->onDelete('cascade');
            $table->timestamps();
        });
    }

    public function down(): void {
        Schema::dropIfExists('books');
    }
};
```

5. create_loans_table.php

```
<?php

use Illuminate\Database\Migrations\Migration;
```

```
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration {
    public function up(): void {
        Schema::create('loans', function (Blueprint $table) {
            $table->id();
$table->foreignId('member_id')->constrained()->onDelete('cascade');
            $table->foreignId('book_id')->constrained()->onDelete('cascade');
            $table->date('loan_date');
            $table->date('return_date')->nullable();
            $table->timestamps();
        });
    }

    public function down(): void {
        Schema::dropIfExists('loans');
    }
};
```

◈ Seeder Files

1. `MemberSeeder.php`

```
<?php

namespace Database\Seeders;

use Illuminate\Database\Seeder;
```

```
use Illuminate\Support\Facades\DB;


class MemberSeeder extends Seeder {

    public function run(): void {

        DB::table('members')->insert([

            ['name' => 'Ali', 'email' => 'ali@mail.com', 'phone' =>
'081234567'],

            ['name' => 'Budi', 'email' => 'budi@mail.com', 'phone' =>
'081987654'],

        ]);

    }

}
```

2. AuthorSeeder.php

```
<?php


namespace Database\Seeders;


use Illuminate\Database\Seeder;

use Illuminate\Support\Facades\DB;


class AuthorSeeder extends Seeder {

    public function run(): void {

        DB::table('authors')->insert([

            ['name' => 'Tere Liye'],

            ['name' => 'Andrea Hirata'],

        ]);

    }

}
```

3. PublisherSeeder.php

```
<?php

namespace Database\Seeders;

use Illuminate\Database\Seeder;
use Illuminate\Support\Facades\DB;

class PublisherSeeder extends Seeder {
    public function run(): void {
        DB::table('publishers')->insert([
            ['name' => 'Gramedia'],
            ['name' => 'Mizan'],
        ]);
    }
}
```

4. BookSeeder.php

```
<?php

namespace Database\Seeders;

use Illuminate\Database\Seeder;
use Illuminate\Support\Facades\DB;

class BookSeeder extends Seeder {
    public function run(): void {
```

```
        DB::table('books')->insert([
            ['title' => 'Bumi', 'author_id' => 1, 'publisher_id' => 1],
            ['title' => 'Laskar Pelangi', 'author_id' => 2, 'publisher_id' =>
2],
        ]);
    }
}
```

5. `DatabaseSeeder.php`

Jangan lupa daftarkan semua seeder di `DatabaseSeeder.php`:

```
<?php

namespace Database\Seeders;

use Illuminate\Database\Seeder;

class DatabaseSeeder extends Seeder {
    public function run(): void {
        $this->call([
            MemberSeeder::class,
            AuthorSeeder::class,
            PublisherSeeder::class,
            BookSeeder::class,
        ]);
    }
}
```

🚀 Jalankan Seeder

```
php artisan migrate:fresh --seed
```